

GENETIC ENGINEERING: GM CROPS AND CRISPR

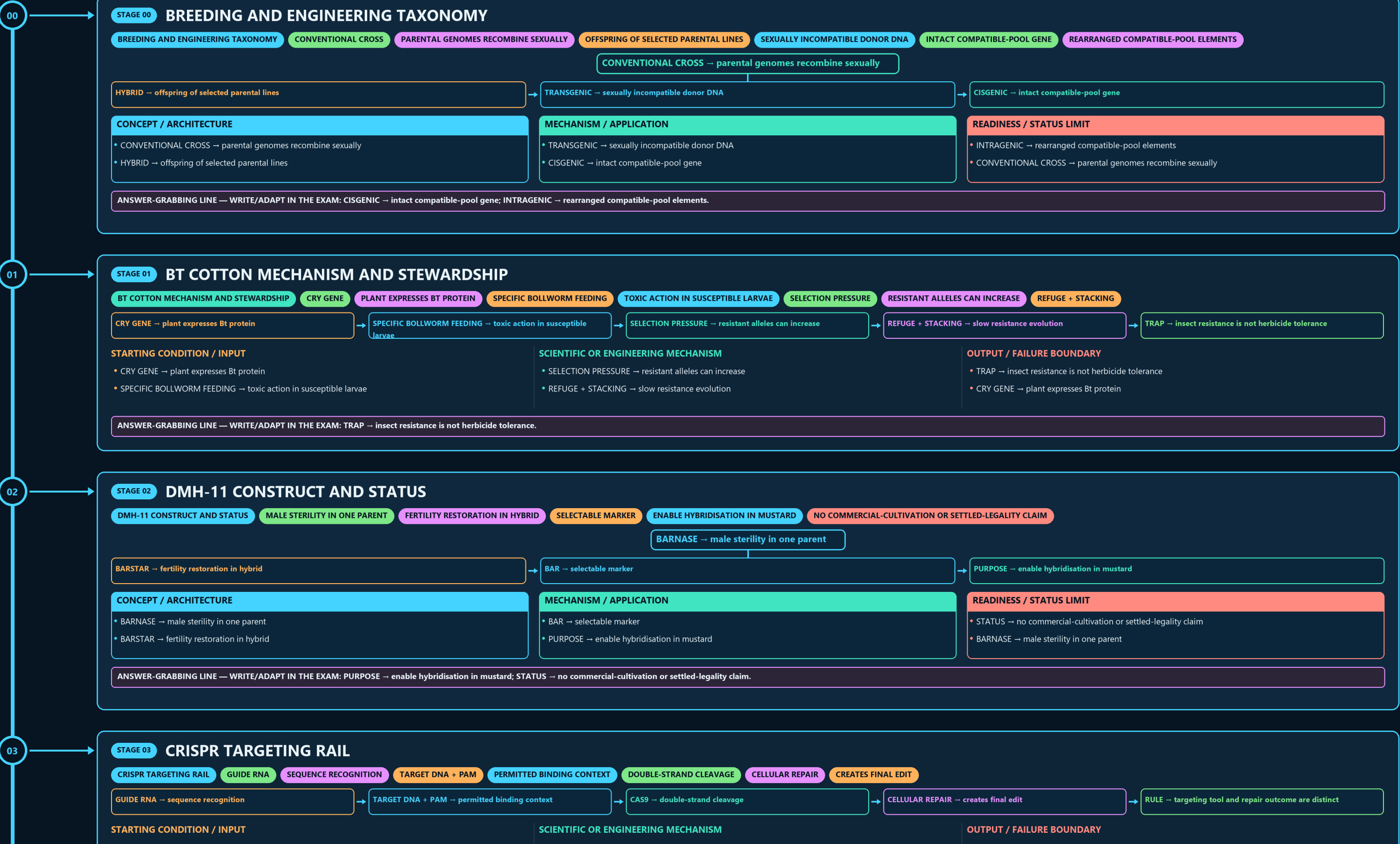
BREEDING AND ENGINEERING TAXONOMY → BT COTTON MECHANISM AND → DMH-11 CONSTRUCT AND STATUS → CRISPR TARGETING RAIL → EDITING-OUTCOME MATRIX → SDN CLASSIFICATION LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PURPOSE → enable hybridisation in mustard; STATUS → no commercial-cultivation or settled-legality claim.

03

STAGE 03 CRISPR TARGETING RAIL

CRISPR TARGETING RAIL GUIDE RNA SEQUENCE RECOGNITION TARGET DNA + PAM PERMITTED BINDING CONTEXT DOUBLE-STRAND CLEAVAGE CELLULAR REPAIR CREATES FINAL EDIT

GUIDE RNA → sequence recognition

TARGET DNA + PAM → permitted binding context

CAS9 → double-strand cleavage

CELLULAR REPAIR → creates final edit

RULE → targeting tool and repair outcome are distinct

STARTING CONDITION / INPUT

- GUIDE RNA → sequence recognition
- TARGET DNA + PAM → permitted binding context

SCIENTIFIC OR ENGINEERING MECHANISM

- CAS9 → double-strand cleavage
- CELLULAR REPAIR → creates final edit

OUTPUT / FAILURE BOUNDARY

- RULE → targeting tool and repair outcome are distinct
- GUIDE RNA → sequence recognition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → targeting tool and repair outcome are distinct.

04

STAGE 04 EDITING-OUTCOME MATRIX

EDITING-OUTCOME MATRIX TEMPLATE-FREE INDELS AND KNOCKOUT DONOR-TEMPLATE SUBSTITUTION OR INSERTION BASE EDITING CHEMICAL BASE CONVERSION WITHOUT DSB PRIME EDITING TEMPLATED SHORT WRITING WITHOUT DSB PRECISION MECHANISMS ARE NOT INTERCHANGEABLE

SYSTEM / DEVICE / INSTITUTION

NHEJ → template-free indels and knockout
HDR → donor-template substitution or insertion

DECISIVE TECHNICAL DISTINCTION

BASE EDITING → chemical base conversion without DSB
PRIME EDITING → templated short writing without DSB

STATUS / UNIT / EVIDENCE LIMIT

TRAP → precision mechanisms are not interchangeable
NHEJ → template-free indels and knockout

SYSTEM / DEVICE / INSTITUTION

- NHEJ → template-free indels and knockout
- HDR → donor-template substitution or insertion

DECISIVE TECHNICAL DISTINCTION

- BASE EDITING → chemical base conversion without DSB
- PRIME EDITING → templated short writing without DSB

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → precision mechanisms are not interchangeable
- NHEJ → template-free indels and knockout

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRIME EDITING → templated short writing without DSB.

05

STAGE 05 SDN CLASSIFICATION LADDER

SDN CLASSIFICATION LADDER NO TEMPLATE; SMALL EDIT SHORT TEMPLATE; DEFINED SMALL EDIT GENE-SIZED TARGETED INSERTION 2022 LIGHTER TRACK SDN-1/2 FREE OF EXOGENOUS DNA FULL SCRUTINY SDN-3 REMAINS OUTSIDE EXEMPTION

SDN-1 → no template; small edit

SDN-2 → short template; defined small edit

SDN-3 → gene-sized targeted insertion

2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

STARTING CONDITION / INPUT

- SDN-1 → no template; small edit
- SDN-2 → short template; defined small edit

SCIENTIFIC OR ENGINEERING MECHANISM

- SDN-3 → gene-sized targeted insertion
- 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

OUTPUT / FAILURE BOUNDARY

- FULL SCRUTINY → SDN-3 remains outside exemption
- SDN-1 → no template; small edit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA.

06

STAGE 06 SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAP RECOMBINANT-DNA ADVICE INSTITUTIONAL DAY-TO-DAY OVERSIGHT RESEARCH AND CONTAINED STAGES ENVIRONMENTAL RELEASE AND LARGE-SCALE USE SBCC + DLC STATE AND DISTRICT MONITORING

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight

RCGM / DBT → research and contained stages

GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE

- RDAC → recombinant-DNA advice
- IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION

- RCGM / DBT → research and contained stages
- GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT

- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

06

STAGE 06

SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAPRECOMBINANT-DNA ADVICERDACCORE GMO BIOSAFETY FRAMEWORKCONTAINED WORKENVIRONMENTAL APPRAISALGM-FOOD SAFETY APPROVAL AND LABELLING TRACKCROP SYSTEMAGRONOMIC EVALUATION AND VARIETY PROCESSES

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight

RCGM / DBT → research and contained stages

GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION

RCGM / DBT → research and contained stages

GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT

SBCC + DLC → state and district monitoring

RDAC → recombinant-DNA advice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SBCC + DLC → state and district monitoring.

07

STAGE 07

LAW AND REGULATOR SPLIT

LAW AND REGULATOR SPLITRULES 1989CORE GMO BIOSAFETY FRAMEWORKCONTAINED WORKENVIRONMENTAL APPRAISALGM-FOOD SAFETY APPROVAL AND LABELLING TRACKCROP SYSTEMAGRONOMIC EVALUATION AND VARIETY PROCESSES

SYSTEM / DEVICE / INSTITUTION

RULES 1989 → core GMO biosafety framework

RCGM → research / contained work

DECISIVE TECHNICAL DISTINCTION

GEAC → environmental appraisal

FSSAI → GM-food safety approval and labelling track

STATUS / UNIT / EVIDENCE LIMIT

ICAR / CROP SYSTEM → agronomic evaluation and variety processes

RULES 1989 → core GMO biosafety framework

SYSTEM / DEVICE / INSTITUTION

RULES 1989 → core GMO biosafety framework

RCGM → research / contained work

DECISIVE TECHNICAL DISTINCTION

GEAC → environmental appraisal

FSSAI → GM-food safety approval and labelling track

STATUS / UNIT / EVIDENCE LIMIT

ICAR / CROP SYSTEM → agronomic evaluation and variety processes

RULES 1989 → core GMO biosafety framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ICAR / CROP SYSTEM → agronomic evaluation and variety processes.

08

STAGE 08

INDIAN DECISION TIMELINE

INDIAN DECISION TIMELINE09 FEB 2010BT BRINJAL MORATORIUM30 MAR 2022SDN-1/2 EXEMPTION OM17 MAY 2022DBT GENOME-EDITED PLANT GUIDELINES23 JUL 2024

09 FEB 2010 → Bt brinjal moratorium

30 MAR 2022 → SDN-1/2 exemption OM

17 MAY 2022 → DBT genome-edited plant guidelines

23 JUL 2024 → Supreme Court DMH-11 split verdict

04 APR 2025 → two genome-edited rice varieties announced

CONCEPT / ARCHITECTURE

09 FEB 2010 → Bt brinjal moratorium

30 MAR 2022 → SDN-1/2 exemption OM

MECHANISM / APPLICATION

17 MAY 2022 → DBT genome-edited plant guidelines

23 JUL 2024 → Supreme Court DMH-11 split verdict

READINESS / STATUS LIMIT

04 APR 2025 → two genome-edited rice varieties announced

09 FEB 2010 → Bt brinjal moratorium

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 23 JUL 2024 → Supreme Court DMH-11 split verdict.

09

STAGE 09

BIOSAFETY MECHANISM MAP

BIOSAFETY MECHANISM MAPTARGET PEST PRESSURERESISTANCE EVOLUTIONSEED MOVEMENTGENE-FLOW EXPOSURETRAIT EXPRESSIONNON-TARGET EXPOSURE PATHWAYSIMILAR DNA SITES

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposure

TRAIT EXPRESSION → non-target exposure pathway

SIMILAR DNA SITES → possible off-target edit

STARTING CONDITION / INPUT

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposure

SCIENTIFIC OR ENGINEERING MECHANISM

TRAIT EXPRESSION → non-target exposure pathway

SIMILAR DNA SITES → possible off-target edit

OUTPUT / FAILURE BOUNDARY

RESPONSE → characterisation, mitigation and monitoring

MUST REMEMBER: Genetic engineering changes DNA through defined tools; GM crops and CRISPR...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Genetic engineering changes DNA through defined tools; GM crops and CRISPR.

10

STAGE 10

FARMER-LIVELIHOOD EVALUATION

